Hypermedia Concepts

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Hypertext

Text that is displayed on an electronic medium.

Contains links to other text in the same document.

Or to external files or other documents.

Defines navigational pathways over the internet.

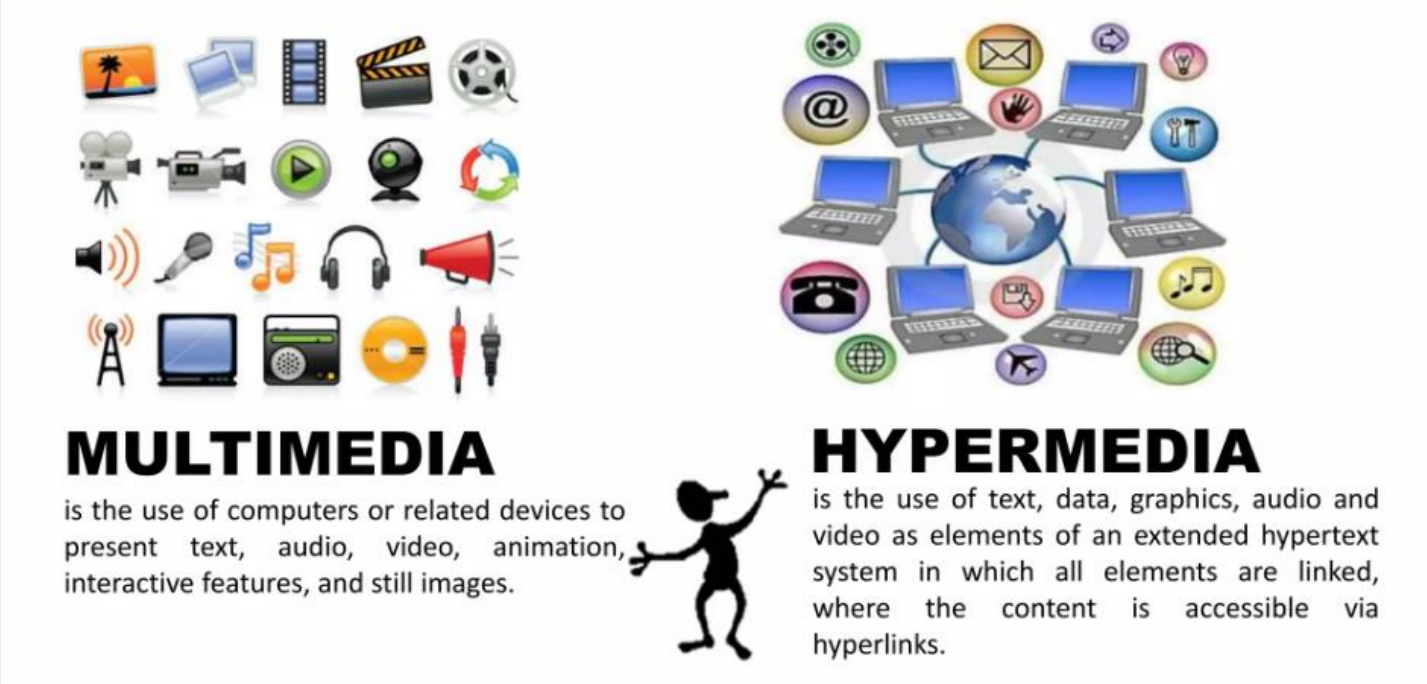
Hypermedia – The Logical Evolution

Extension of Hypertext

Use non-textual media to create hyperlinks

Audio, video, image, animation, graphics

Also called as interactive multimedia

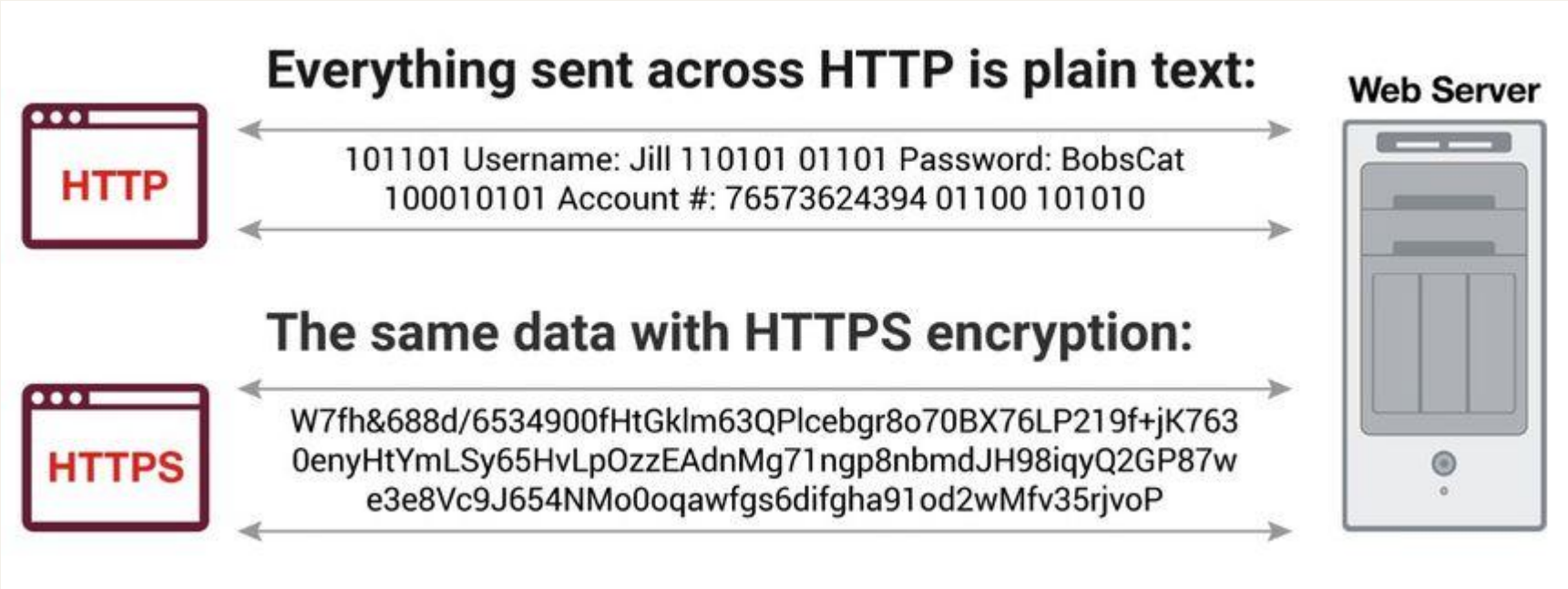


Markup

- Everything in a document that is not content.
- Markup indicators are called tags.
- **Procedural Markup**
  - Unique to a specific software package.
  - Eg: <Ctrl>+B for making text bold in MS Word.
- **Descriptive Markup**
  - Recognisable by different application softwares.
  - Eg: a tag used for making text bold in HTML.
  - Interpreted by different browsers, and they produce the actual change in their own way.
- **Markup language:** syntax or system which uses markup rules to define document format and structure.
- Eg: HTML and XML

Hypertext Transfer Protocol

- Set of rules for **data communication** in a hypermedia information system.
- Functions in a request-response manner.
- **Stateless:** server doesn't retain information about a specific client.
- HTTP connections can be made secure by using Secure Sockets Layer (SSL) → HTTPS



HTTP Status Codes

1xx Information Exchange	100	Continue	The server has received the request headers and the client should proceed to send the request body.
	101	Switching Protocols	The requester has asked the server to switch protocols and the server has agreed to do so.
2xx Success	200	OK	Standard response for successful HTTP requests.
	201	Created	The request has been fulfilled, resulting in the creation of a new resource.
	202	Accepted	The request has been accepted for processing.
	204	No Content	The server processed the request, but is not returning any content.
3xx Redirection	304	Not Modified	Indicates that the resource has not been modified. In such a case, there is no need to retransmit the resource since the client still has a previously-downloaded copy.
	305	Use Proxy	The requested resource is available only through a proxy.
	307	Temporary Redirect	In this case, the request should be repeated with another URI; however, future requests should still use the original URI.
	308	Permanent Redirect	This and all future requests should be directed to the given URI.

HTTP Status Codes

4xx Client Errors	400	Bad Request	The server cannot or will not process the request due to an apparent client error.
	401	Unauthorised	Authentication is required but has failed or has not yet been provided.
	403	Forbidden	The request contained valid data and was understood by the server, but the server is refusing action.
	404	Not Found	The requested resource could not be found.
	408	Request Timeout	The server timed out waiting for the request.
	414	URI Too Long	The URI provided was too long for the server to process.
5xx Server Errors	500	Internal Server Error	A generic error message, given when an unexpected condition was encountered and no more specific message is suitable.
	502	Bad Gateway	The server was acting as a gateway or proxy and received an invalid response from the upstream server.
	503	Service Unavailable	The server can't handle the request. Generally, this is a temporary state.
	504	Gateway Timeout	The server was acting as a gateway or proxy and did not receive a timely response from the upstream server.

Document Type Definition

- Set of rules which validate a markup language.
- Main building blocks:
  - **Elements:** represented by tags in HTML; use them to describe different parts of a data.
  - **Attributes:** provide additional information about tags, located inside opening tag of an element; use them to include metadata
  - **Entities:** set of characters having special meaning (eg: nbsp)
  - **PCDATA:** parsed character data → text found between start and end tags, but without any entities.
  - **CDATA:** character data → not parsed by parser, represents text inside tags

```
<!ELEMENT element-name (element-content)>
<!ATTLIST element-name attribute-name attribute-type default-value>

<!ELEMENT html (head, body)>
<!ELEMENT p (#PCDATA | p | ul | dl | table | h1|h2|h3)*>
```

Document Type Definition

```
<students>
    <student>
        <id>...</id>
        <name>...</name>
        <department>...</department>
        <grades>
            <subject>
                <name>...</name>
                <score>...</score>
            </subject>
        </grades>
    </student>
</students>
```

<!ELEMENT students (student+)>
<!ELEMENT student (id, name, department, grades)>
<!ELEMENT id (#PCDATA)>
<!ELEMENT name (#PCDATA)>
<!ELEMENT grades (subject+)>
<!ELEMENT subject (name, score)>
For an entry <score format=".4f"/>, the corresponding DTD declaration can be <!ATTLIST score format CDATA ".10f">

Standard Generalised Markup Language

- Prescribes a standard format for embedding descriptive markup within a document. Also specifies a standard method for describing the structure of a document.
- Three Parts:
  - Structure: DTD describes structure
  - Content: The actual information
  - Style: Two standards-based style sheets
    - FOSI: Formatting Output Specification Instance
    - DSSSL: Document Style Semantics and Specification Language

```
<!ELEMENT    - -    students (student+)>
<!ELEMENT    - -    student (id, name, department, grades)>
<!ELEMENT    - O    id (#PCDATA)>
<!ELEMENT    - O    grades (subject+)>
<!ELEMENT    - O    subject (name, score)>
```

Dynamic HTML

- **Combining HTML, CSS, and JS** to create animated web sites.

HTML	Predominant markup scripting language for WWW; based on SGML
CSS	Allows for sophisticated sizing, positioning, and other appearance related manipulations of web page contents
JS	Cross-platform, object-based scripting language for client (user interactions) and server (database) applications

Extensible Markup Language

HTML	focuses on how the content on a page is displayed to the user
XML	concerned with how to structure and store data, such that it can subsequently be searched for and transported

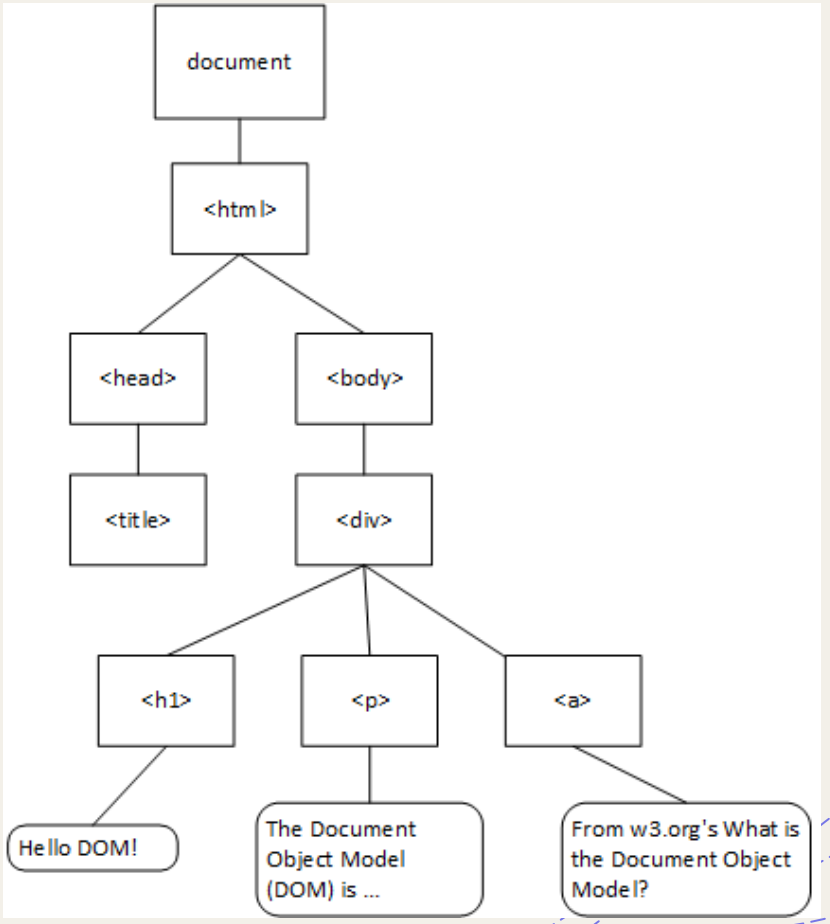
- Just information wrapped in tags.
- XML does not have standard tags. Rather, it enables the user to create customised tags to suit the document being stored.
- XML Schema – an alternative to DTD for XML.

Extensible HTML

- XHTML – a family of markup languages generally regarded as extensions of HTML.
- Based on XML.
- **Why?:** Provided a way for developers to migrate to the XML world while maintaining compatibility with the older HTML platform.
- But, XHTML working group **officially expired in 2007** after it was decided that HTML 5 was more suitable as the next-gen HTML.
- XHTML 5.0 – latest version, developed as part of HTML 5 specifications instead of a separate standard.

Document Object Model

- Underlying platform to enable **manipulation of web objects** in a language independent manner.
- Used by languages like JS to change content of a page **dynamically**.
- 3 different DOM models – DOM by W3C, IE specific DOM, Netscape DOM.



Document Object Model - HTML

HTML Document
<html>   <head>   <title>My Website</title>   </head>   <body>   <h1>Welcome to DOM</h1>   This is my website.   </body> </html>

Corresponding DOM Tree

- Document (root)
  - html
  - head
  - title
  - "My Website"
  - body
  - h1
  - "Welcome"
  - p
  - "This is my website."

Document Object Model – XML

XML Document Snippet

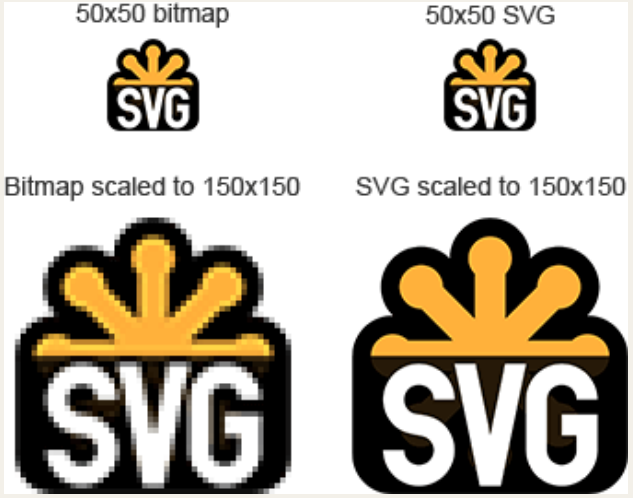
```
<student>
    <id>2022506001</id>
    <name>K Mohamed Jamal Basha </name>
    <department>IT</department>
    <grades>
        <subject>
            <name>MMT</name>
            <score>100</score>
        </subject>
    </grades>
</student>
```

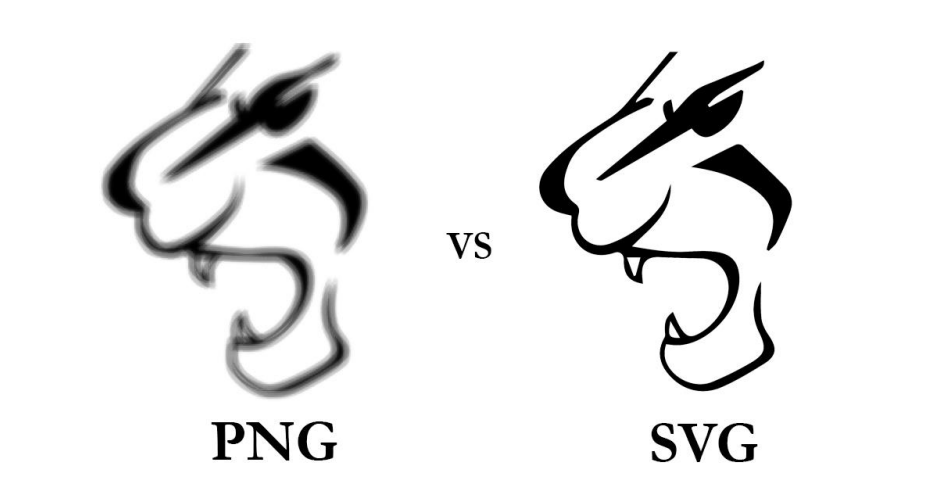
Corresponding DOM Tree

```
- …
  - student
      - id
          - "2022506001"
      - name
          - "K Mohamed Jamal Basha"
      - department
          - "IT"
      - grades
          - subject
              - name
                  - "MMT"
              - score
                  - 100
```

Scalable Vector Graphics

- Set of XML-based specifications for describing 2D vector graphics.
- Being vector formats, they don't suffer from **pixelation** due to zooming.
- Being XML, they can be **edited** using a text editor, can be indexed.
- Perfect for **simple graphics** and illustrations.
- Not the best choice for more complex images, which contain a large number of small elements - SVG files can grow significantly in size in such cases.

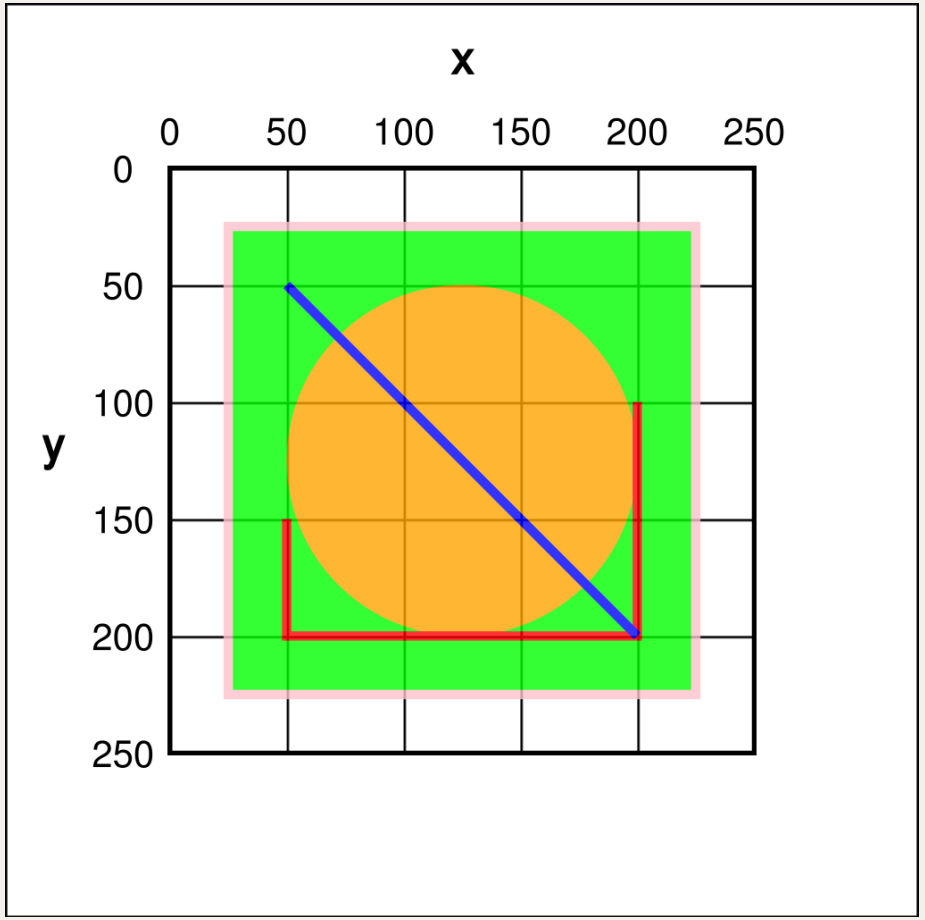




Scalable Vector Graphics

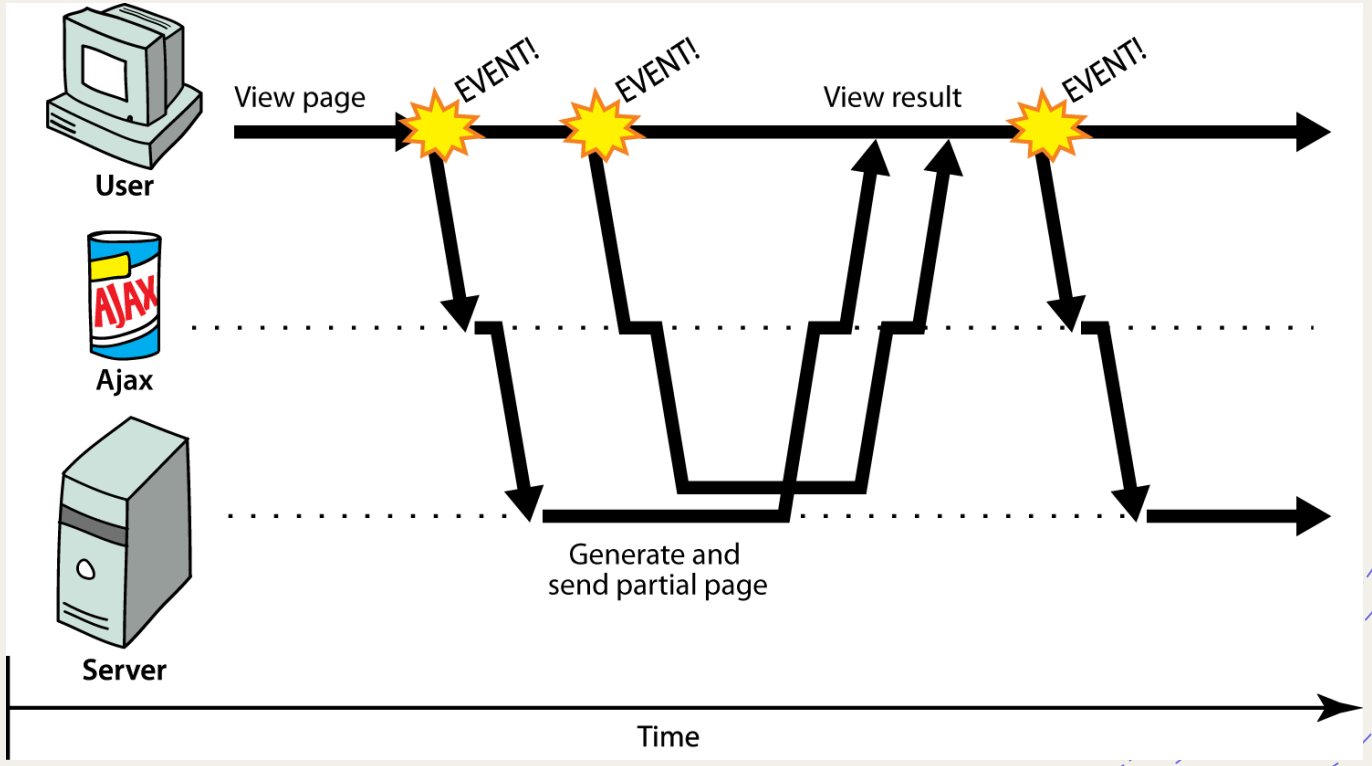
```
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<!DOCTYPE svg PUBLIC "-//W3C//DTD SVG 1.1//EN"
"http://www.w3.org/Graphics/SVG/1.1/DTD/svg11.dtd">
<svg width="391" height="391" viewBox="-70.5 -70.5 391 391" xmlns="http://www.w3.org/2000/svg"
xmlns:xlink="http://www.w3.org/1999/xlink">
<rect fill="#fff" stroke="#000" x="-70" y="-70" width="390" height="390"/>
<g opacity="0.8">
        <rect x="25" y="25" width="200" height="200" fill="lime" stroke-width="4" stroke="pink" />
        <circle cx="125" cy="125" r="75" fill="orange" />
        <polyline points="50,150 50,200 200,200 200,100" stroke="red" stroke-width="4" fill="none" />
        <line x1="50" y1="50" x2="200" y2="200" stroke="blue" stroke-width="4" />
</g>
</svg>
```

Scalable Vector Graphics



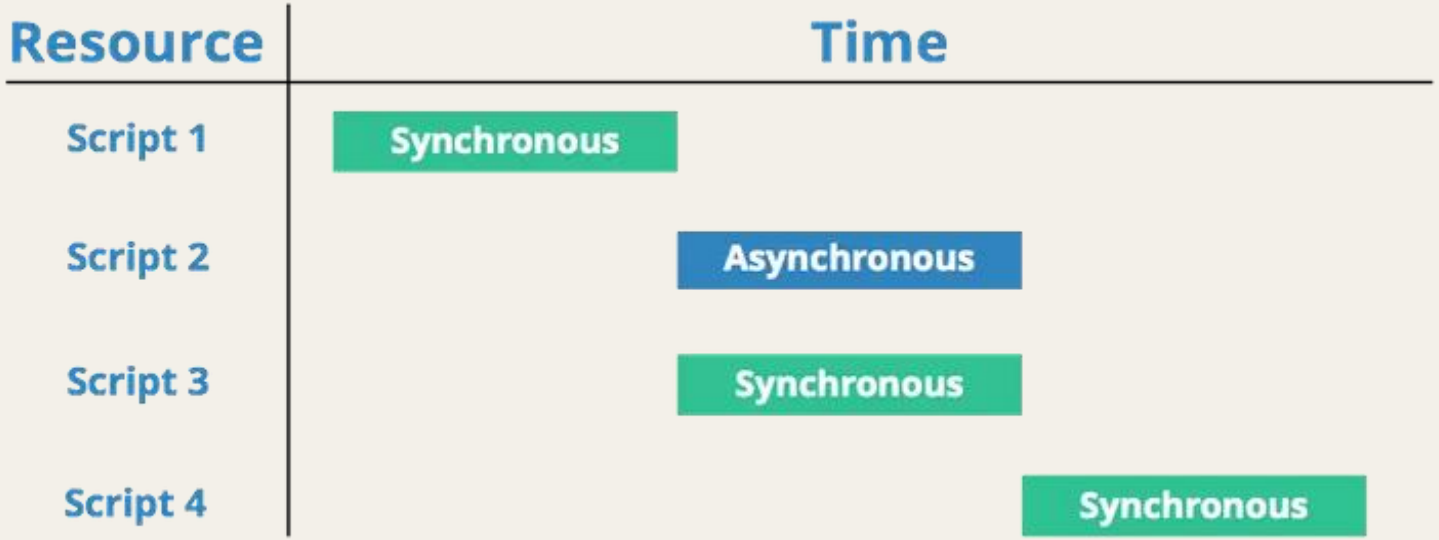
Asynchronous JavaScript and XML

- AJAX – Set of techniques used on client side to **retrieve data** from the server **asynchronously** in the background.
- This means the rest of the page content is undisturbed and can still be interacted with.



Asynchronous JavaScript and XML

There are 3 synchronous scripts being loaded one after another. However, there is also an asynchronous script which can be loaded at the same time as the synchronous script, which makes for a much **more efficient loading process**.



Multipurpose Internet Mail Extension

- MIME – **identifier** for file formats on the Internet.
- Originally, for use with email sent through SMTP, but later expanded to other protocols like HTTP and RTP.
- In HTTP, servers insert a **MIME header field** at the beginning of any Web transmission. Clients use the content type or media type header to select an appropriate viewer application for the type of data indicated.

Multipurpose Internet Mail Extension

- MIME Header contains
  - Version
  - Media Type: composed of two parts, a **type** and a **sub-type**.
  - Media Disposition
    - Inline: automatically displayed when the message is displayed
    - Attachment: requires some user action, like download
  - Encoding: mentions the encoding schemes used for the media.

Text	Image	Audio	Application	Video
text/plain	image/gif	audio/mpeg	application/javascript	video/mpeg
text/html	image/jpeg	audio/mp4	application/pdf	video/mp4
text/css	image/png	audio/ogg	application/zip	video/ogg
text/xml	image/tiff	audio/wave	application/powerpoint	video/quicktime

Synchronised Multimedia Integration Language

- XML-based W3C recommendation for **describing multimedia presentations**.
- Defines markup elements for timing, layout, animations, and transitions.
- **Core of SMIL:** Synchronizing zones with multimedia content.
- Divides the different media components into separate files and streams and then displays them together as if they were a single multimedia stream.
- Eg: Delivering captions/audio in a synchronized manner with video.

Synchronised Multimedia Integration Language

- A snippet of an SMIL file defining a slide change sequence.
- Defines slide ID, the image file name for the slide, start time, end time, and type of slide.

```
<slidechange>
        <slide id="1" imagefile="Slide001.bmp" st="0.0000" et="5.0400" type="normal" />
        <slide id="2" imagefile="Slide002.bmp" st="5.0400" et="5.5200" type="skip" />
        <slide id="3" imagefile="Slide003.bmp" st="5.5200" et="5.9600" type="skip" />
        <slide id="4" imagefile="Slide004.bmp" st="5.9600" et="6.2800" type="skip" />
        <slide id="5" imagefile="Slide005.bmp" st="6.2800" et="7.2400" type="skip" />
        <slide id="6" imagefile="Slide006.bmp" st="7.2400" et="12.8400" type="normal" />
        <slide id="7" imagefile="Slide007.bmp" st="12.8400" et="12.9600" type="skip" />
        <slide id="8" imagefile="Slide008.bmp" st="12.9600" et="13.1200" type="skip" />
        <slide id="9" imagefile="Slide009.bmp" st="13.1200" et="13.2800" type="skip" />
        <slide id="10" imagefile="Slide010.bmp" st="13.2800" et="13.4800" type="skip" />
        <slide id="11" imagefile="Slide011.bmp" st="13.4800" et="19.2000" type="normal" />
        <state TotalSlide="11" StablePeriod="3" UnstablePeriod="2" />
</slidechange>
```

Multimedia Messaging Service

- Technique for sending messages containing multimedia content to and from mobile devices.
- MMS messages can contain text (formatted with different fonts, colours, and styles), images, audio, video, and animations.
- MMS is a natural evolution of SMS, but MMS is still limited.
- **Rich Communication Services (RCS):** a newer standard which
  - provides end-to-end encryption,
  - enables sharing of high-resolution photos and videos,
  - Provides typing indicators
  - receive read receipts.

SMS vs MMS vs RCS

Feature	SMS	MMS	RCS
Character Limit	160 characters	1600 characters	No
File Size Limit	No file support	5 MB	100 MB
Format	Text	Text and Multimedia	Text and Multimedia
Sent Via	Cellular	Cellular	Wi-Fi or Cellular
Security	No encryption	No E2EE	End-to-End Encrypted
Read Receipts	No	No	Yes
Typing Indicators	No	No	Yes
Group Chat Limit	No support	10 people	100 people
Internet Required	No	No	Yes
Reactions/Emojis	Limited (text-based)	Limited	Yes
Voice Messages	No	No	Yes
Video Calls	No	No	Yes
Backup Support	No	No	Cloud Backup

Wireless Application Protocol

- Open standard protocol suite to display Internet content on mobile devices. Has become obsolete, due to advancements in modern mobile browsers.

Enables delivery of web-based content.	Application Layer (WAE)
Responsible for maintaining a session between the wireless device and the server.	Session Layer (WSP)
Provides a reliable transport mechanism for transmitting data between the wireless device and the server.	Transaction Layer (WTP)
Provides public-key encryption based security, similar to SSL.	Security Layer (WTLS)
Helps in transport of data in an unreliable way, similar to UDP.	Transport Layer (WDP)

Thank You

Go Back